4. Alert Triage with Automation

1. Objective

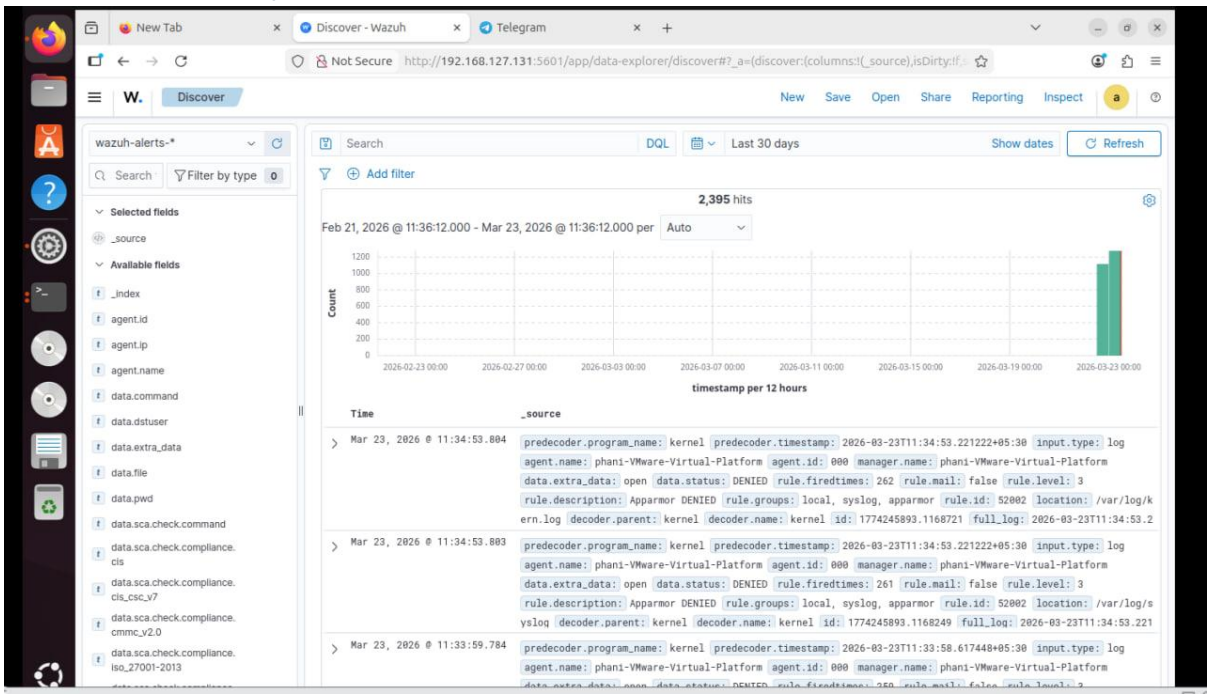
To simulate alert triage using Wazuh and automate threat validation using TheHive integrated with VirusTotal via Cortex.

2. Tools Used

1. Wazuh (SIEM for alert generation)
2. TheHive (Incident Response Platform)
3. Cortex (Analyzer Engine)
4. VirusTotal (Threat Intelligence)

3. Triage Simulation

A mock alert was generated in Wazuh representing a suspicious file download activity.

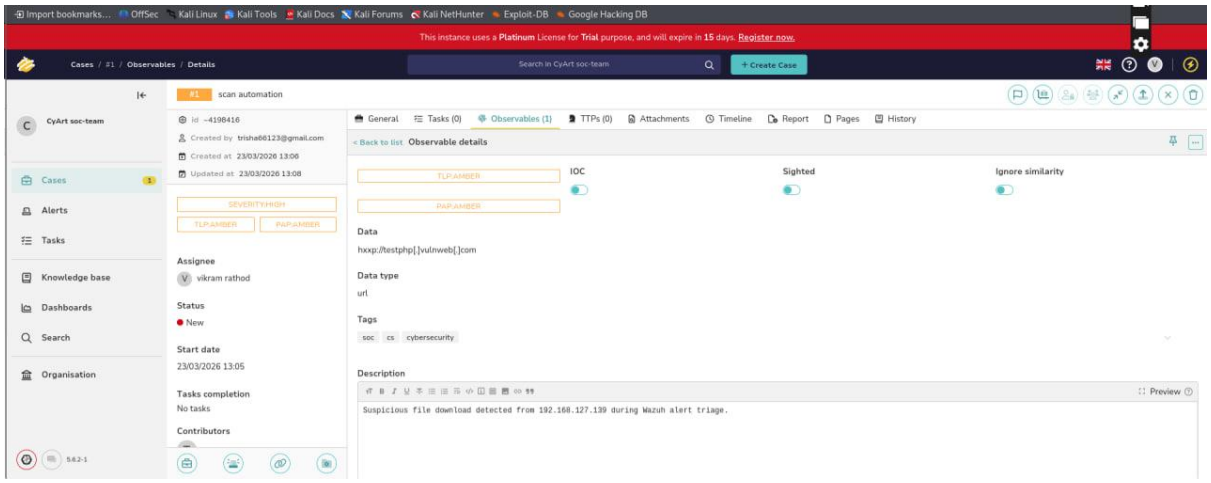


The alert indicated a potential malicious file download originating from an internal host. The event was reviewed in Wazuh dashboard and escalated for further investigation.

	A	B	C	D	E	F	G
1	Alert ID	Description	Source I	Priority	Status		
2	005	File Download	192.168	High	Open		
3							
4							

4. Case Creation in TheHive

A case was created in TheHive for alert ID 005
Observable added:
Type: URL / File Hash
Value: Suspicious download indicator
Tags applied: "malware", "suspicious-download", "wazuh-alert"
TLP: AMBER
PAP: AMBER

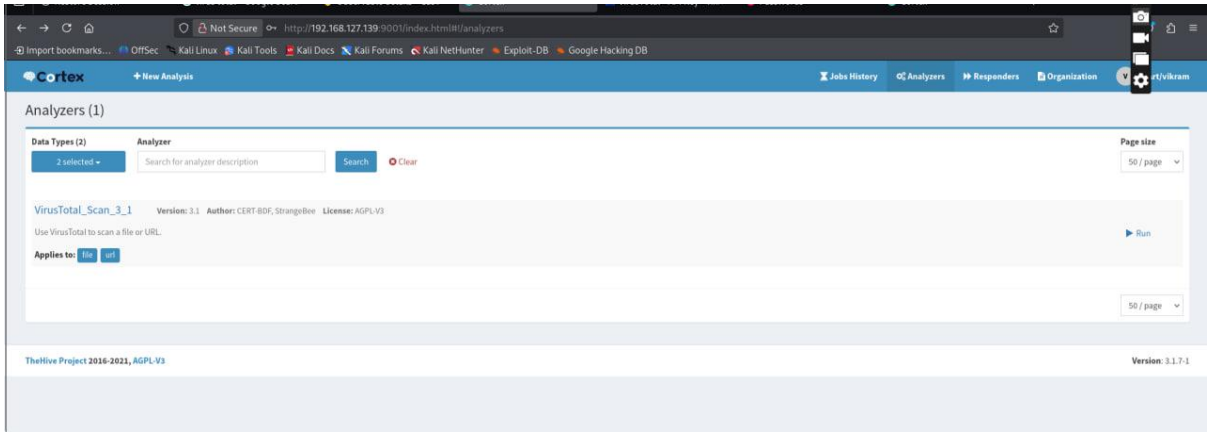


5. Automated Validation using VirusTotal

Cortex was integrated with TheHive and configured with VirusTotal API key.
The VirusTotal_Scan_3_1 analyzer was executed on the observable.

Steps:

1. Open observable in TheHive
2. Run analyzer → VirusTotal_Scan_3_1
3. Submit hash/URL for scanning



6. Results Summary (50 Words)

The alert was validated using VirusTotal integration in TheHive. The file hash was automatically analyzed across multiple antivirus engines. Several detections confirmed malicious behavior. Automation improved triage efficiency by eliminating manual checks and accelerating decision-making using real-time threat intelligence correlation.

7. Outcome

- Successfully simulated alert triage workflow
- Automated enrichment using VirusTotal
- Reduced manual analysis effort
- Demonstrated SOC-level incident response pipeline

8. Conclusion

This implementation showcases effective integration of SIEM and SOAR tools for automated threat validation. The workflow enhances

detection accuracy, speeds up response time, and aligns with real-world SOC operations.